

TARUN CHOWDALA

Full Stack Developer | React.js | FastAPI | Node.js
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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

- Full Stack Developer with **2+ years** of experience building, maintaining, and scaling production-grade web applications using **React.js**, **FastAPI**, **Node.js**, **PostgreSQL**, **Redis**, and **GCP**.
- Experienced in developing end-to-end features, designing **REST APIs**, and maintaining distributed systems involving queues, background jobs, and real-time communication.
- Hands-on experience with **LangChain**, **RAG** pipelines, and **vector databases** for AI-powered applications; actively building deeper GenAI expertise.
- Comfortable taking ownership of frontend and backend systems, debugging complex production issues, and delivering in fast-paced **Agile** environments.

TECHNICAL SKILLS

Frontend: React.js, Context API, Zustand, HTML5, CSS3, Bootstrap, Responsive Design

Backend: FastAPI (Python), Node.js, Express.js

Databases: PostgreSQL, MongoDB, Redis

AI / GenAI: LangChain, LangGraph, AI Agents, RAG, Vector Databases (FAISS)

Cloud & Tools: Google Cloud Platform (GCP), Docker, Hostinger, Git, GitHub, Postman, VS Code

Languages: JavaScript (ES6+), TypeScript, Python

WORK EXPERIENCE

Swara Tech

Sep 2024 – Present

Full Stack Developer | Hyderabad, India — ContactSwing (AI-driven SaaS Platform: Calls, Chat, Email & Automation)

- Built **15+ production-grade screens** and developed **50+ reusable React.js components**, improving code reusability and accelerating feature delivery across the platform.
- Took ownership of frontend architecture by organizing folder structure, standardizing component patterns, and improving maintainability across multiple modules.
- Implemented global state management using **Context API** and **Zustand**, ensuring consistent state handling across complex multi-feature interfaces.
- Optimized frontend performance using **React Query** for caching, background synchronization, and request deduplication, reducing unnecessary API calls and improving user experience.
- Developed and maintained **REST APIs** using **FastAPI** (Python) and **Node.js** to support production features across voice, chat, email, and automation modules.
- Designed and maintained **PostgreSQL** database schemas and queries, ensuring reliable and efficient data operations across platform modules.
- Built and enhanced real-time features using **WebSockets**, enabling low-latency communication for chat-based workflows.
- Worked on asynchronous workflows involving cron jobs, **queue workers**, **Redis**, and **Google Cloud Tasks** to support background processing and campaign automation.

- Currently act as the primary full-stack developer responsible for maintaining and enhancing multiple production applications, including frontend systems, **4 backend services**, and **GCP**-hosted infrastructure.
- Debugged and resolved complex production issues by analyzing application logs, **Docker** container logs, and API traces, improving system reliability and reducing service disruptions.
- Used **TypeScript** with React.js, writing interfaces for API responses and typing variables to improve code reliability.
- Collaborated with cross-functional teams in an Agile environment, participating in sprint planning, code reviews, and feature discussions to deliver production-ready solutions.

Instedia

May 2024 – Jul 2024

Frontend Developer Intern | Hyderabad, India

- Developed and delivered reusable, responsive UI components using **React.js**, HTML5, CSS3, and Bootstrap, accelerating feature delivery.
- Integrated frontend application with backend REST APIs, enabling dynamic data rendering and real-time UI updates.
- Identified and resolved UI bugs and cross-browser compatibility issues, improving application stability.
- Applied **Git** and **GitHub** for version control, branching, and collaborative code reviews within an Agile team environment.

KEY PROJECTS

AI Web Search Agent (Scout)

Tech Stack: Python, Streamlit, LangChain, LangGraph, Groq, Serper API, Wikipedia API, YouTube Search

- Developed an AI-powered web search agent using **LangChain** `create_agent` with **LangGraph** **MemorySaver** for multi-turn conversational memory.
- Integrated multiple external tools including Google Web Search, Google News, Google Images, Wikipedia, and YouTube Search, enabling the agent to dynamically choose the appropriate source for user queries.
- Implemented tool-calling workflows where the LLM autonomously selects, executes, and synthesizes information from multiple data sources into coherent responses.
- Added support for multiple **Groq**-hosted LLMs (GPT-OSS 120B, Qwen 3.6 27B, Llama 3.3 70B) to compare reasoning quality, speed, and tool-calling performance.
- Built a **Streamlit** chat interface with model selection, conversation history, image gallery rendering, and clickable YouTube previews.
- Designed modular components separating UI, agent orchestration, prompts, utilities, and tools, improving maintainability and extensibility.

SmartChat AI Platform

Tech Stack: React.js, FastAPI, Python, LangChain, FAISS, HuggingFace Embeddings

- Built a full-stack AI platform as a personal project featuring real-time chat, **RAG**-powered document Q&A, and AI resume tools.
- Built a **RAG pipeline** using **LangChain** and **FAISS** **vector database** with **HuggingFace Embeddings** for semantic document search.
- Built specialized prompt pipelines for AI resume analysis and generation, with multi-turn conversational context handling via **LangChain**.
- Developed all backend APIs using **FastAPI**, delivering high-performance endpoints for AI inference and document processing.

Madivra E-Commerce Platform

Tech Stack: React.js, Node.js, Express.js, MongoDB, Docker, Razorpay, Shiprocket, SMTP, Fast2SMS

- Developed a full-stack e-commerce platform with a feature-rich Admin Dashboard for managing products, categories, discount coupons, inventory, customer orders, and day-to-day business operations.
- Implemented secure OTP-based authentication using Email (**SMTP**) and Mobile (**Fast2SMS**), enabling seamless user registration and login with multi-channel verification.
- Built a complete checkout and payment workflow by integrating the **Razorpay** Payment Gateway, including webhook-based payment verification to ensure reliable order processing even when users disconnect during payment.
- Integrated **Shiprocket** APIs for automated shipment creation, AWB generation, courier assignment, and order fulfillment; implemented Shiprocket webhooks to synchronize real-time delivery status for live order tracking.
- Containerized the application using **Docker** and deployed on **Hostinger VPS**, configuring the production environment for reliable hosting and deployment.
- Configured **SMTP** email services for OTP verification, transactional emails, and order-related notifications.
- Designed and developed **RESTful APIs** for authentication, product management, order processing, payment handling, and shipping workflows with modular and maintainable backend architecture.

CERTIFICATIONS & EDUCATION

Full Stack Development – NxtWave Distributive Technologies | Python Programming – HackerRank

Bachelor of Science in Computer Science — HPN Degree College, Srikakulam

2023